

Question:

A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

$(A) \ 4 + \sqrt{185} \quad (B) \ 16 \quad (C) \ 17 \quad (D) \ 18$
 $\quad (E) \ \sqrt{32} + \sqrt{137}$

Reasoning + Answer:

<think>

1. Understand the problem:

- The cowboy is 4 miles south of the stream .
- The cowboy is also 8 miles west and 7 miles north of his cabin .
- The shortest distance he can travel to water his horse at the stream and return home .

2. Visualize the situation:

- Let the stream be a straight line running east-west .
- The cowboy's starting point (south of the stream) can be considered a point , say (P) .
- The cabin is located at some point (not important for distance calculation).
- He must go to the stream (point (S)), which is 4 miles south .
- He can then go back home , which we'll find after some calculations .

3. Convert distances and straight lines:

- He is 8 miles west and 7 miles north to his cabin .
- Place a coordinate system with point (O) (origin) at the stream , (P) at 4 miles south , (Q) at $(0 , - 4)$, (M) at $(0 , - 4 + 7) = (0 , 3)$.
- He needs to cover 8 miles East and 7 miles North from (Q) to the cabin (M) , a journey $(QM = \sqrt{8^2 + 7^2} = \sqrt{64 + 49} = \sqrt{113})$.

4. Calculate minimum distance:

- He will retrace his steps back to the stream .
- He goes back directly with optimal shortest route with (R) being the one-step ab-initio he employs :
- The stream as a condition creates a piece wise straight est with minimal issue when traced .

5. Distance Calculations:

- Consider actually :
- The optimal estimated retrace (coordinate-frame-calculated simplest) straight est $(Q \to S \to R)$.
- Takes T-shaped calculation , finally state :
- Final simplified by leg . By summing he travels $(\sqrt{185})$, inestimably by tighter (step).

6. summation tie up simplified answer / notation return* calculations sum found* option via $(R \big)$.

In terse simplified consideration of varie er streaming ton matched , accurately summing options for direct *` shortcut*(approx)); Various calculation in = \t adjust dit actors \ ,

<answer> $4 + \sqrt{185}$ </answer>